

Pitch-Tracking Risk Factors and Warning Signs for Shoulder Capsulolabral Injuries in Major League Baseball Pitchers

Morgan R. Dillon,* BS , Michael A. Mastroianni,[†] MD , Nicholas Frappa,* DPT, MS , Danil Chernov,* MA , Andrew J. Luzzi,[†] MD , Matthew G. Alben,* DO , James Muscat,[‡] BS , Kristen F. Nicholson,[§] PHD , Frank J. Alexander,[†] MS, ATC, Robert Ablove,* MD, and Christopher S. Ahmad,^{†||} MD

Investigation performed at Columbia University, New York City, NY, USA

Background: Shoulder capsulolabral injuries are a major source of disability in Major League Baseball (MLB) pitchers. The introduction of Statcast pitch tracking now enables population-level analysis of shoulder injury mechanisms.

Purpose/Hypothesis: The purpose of this study was to identify preinjury kinematic risk factors for arthroscopic capsulolabral repair. It was hypothesized that pitchers who sustain capsulolabral injuries show preinjury kinematic deviations detectable on Statcast data.

Study Design: Case-control study; Level of evidence, 3.

Methods: MLB pitchers undergoing primary capsulolabral repair between 2017 and 2024 ($n = 27$) were matched 1:2 to uninjured controls ($n = 54$) by age, role, handedness, season, and pitch count. Statcast pitch-level data were analyzed across 2 consecutive seasons preceding surgery, yielding 161,979 pitches. Six kinematic variables (release speed, spin rate, arm angle, extension, and horizontal and vertical release position) were extracted across 7 pitch types. To identify potential risk factors and preinjury warning signs, 3 complementary analyses were performed. First, a baseline analysis compared mean kinematic profiles and within-outing variability during the preinjury season (season 1) to characterize potential inherent risk profiles. Second, a longitudinal analysis evaluated change scores across 3 intervals including within season 1, across the winter offseason, and within the injury season, by comparing the first 3 versus last 3 qualifying outings of each period to assess potential progressive trends preceding surgery. Third, to identify acute warning signs, a separate analysis modeled linear trends across the final 5 outings immediately preceding injured list placement. Analyses used linear mixed-effects models with clustered standard errors.

Results: At baseline (season 1), cases demonstrated significantly higher mean velocity and spin, and lower variability in arm angle and release speed, compared with controls across pitch types. On 4-seam fastballs and sinkers, this corresponded to approximately $+1.46$ mph and $+1.57$ mph higher velocity ($P = .009$ and $P = .006$, respectively). Standardized effects showed higher spin rate ($\beta = +0.32$ SD; $P = .02$) and lower variability in arm angle ($\beta = -0.58$ SD; $P = .005$) and release speed ($\beta = -0.37$ SD; $P = .007$). During the injury season (season 2), cases showed a significantly greater decline in velocity on fastballs (-0.68 mph; $P = .009$) and breaking balls (-1.07 mph; $P = .003$). In the final 5 outings before surgery, cases demonstrated a progressive, linear loss on fastball velocity (-0.20 mph/outing) and breaking ball velocity (-0.30 mph/outing; both $P < .001$), while other kinematic measures remained stable.

Conclusion: MLB pitchers who required arthroscopic capsulolabral repair demonstrated a pitch profile characterized by higher velocity and spin coupled with reduced pitch-to-pitch variability. Progressive fastball and breaking ball velocity decline was the clearest marker of impending injury, highlighting a measurable preinjury signature that may inform early detection and intervention.

Keywords: baseball; shoulder; labrum; biomechanics; pitching mechanics; Statcast

Shoulder injuries, particularly those involving the glenoid labrum and capsule, are a leading cause of disability and a significant threat to the careers of baseball pitchers.⁴ Epidemiological studies indicate the shoulder is the most frequently injured site in professional pitchers, accounting for 17% of all injuries and resulting in substantial time lost from competition.⁷ Pitchers suffer a higher incidence of upper extremity injury than position players³⁹ and consequently spend more days on the injured list (IL).¹² Beyond individual burden, the economic impact is considerable, as superior labrum anterior-posterior (SLAP) tears alone cost Major League Baseball (MLB) teams more than \$3.5 million per year.⁴⁰ The prognosis for a successful return to play following labral surgery is notoriously poor, typically ranging from 40% to 62%.^{14,41} Furthermore, many of those who do return never regain their prior performance levels, and rehabilitation is often prolonged, lasting anywhere from 9 to 18 months.¹⁸

These injuries are thought to arise from the extreme biomechanical loads and overuse inherent to MLB pitching, concentrated during the late cocking, early acceleration, and deceleration phases of the throwing motion.^{2,13,37} During the late cocking phase, the arm reaches maximal external rotation (often >170°) and abduction, generating shear forces of approximately 400 N across the glenohumeral joint in the position of anterior shoulder instability.^{9,31} During the subsequent acceleration phase, shoulder internal rotation angular velocity can exceed 7000 deg/s and put increased stress on the posterior labrum.⁴⁶ Immediately after ball release, the deceleration phase requires rapid eccentric muscle contractions to decelerate the arm, placing excessive tensile stress on the biceps anchor and superior labrum.³⁴ In elite throwers, this chronic microtrauma leads to labral and SLAP tears along with attenuation of the surrounding capsule.⁵

The emergence of in-stadium player tracking systems, most notably MLB's Statcast, has revolutionized baseball by providing a rich source of high-frequency, in-game data for players, teams, and the public. This technology, which was first introduced to MLB stadiums in 2015, utilizes Doppler radar and high-speed cameras to capture granular, pitch-by-pitch measurements of key release-adjacent kinematics, enabling detailed pitch-tracking analysis at a population level.³²

Despite the clinical and economic significance of shoulder pathology, contemporary pitch-tracking risk factor research has concentrated on elbow ulnar collateral

ligament (UCL) injuries.^{6,22,28,29,30,38,43} A robust body of literature has leveraged large, public databases to identify risk factors for UCL injury, such as workload, velocity, strike-zone command, pitching ability, and other pitch-specific kinematic patterns.^{6,22,28,29,30,38} In stark contrast, the shoulder has remained significantly understudied within this new analytical framework. While foundational biomechanical studies have elucidated the forces acting on the shoulder, large-scale epidemiological investigations identifying preinjury risk factors for shoulder instability requiring surgery remain lacking.

Therefore, the purpose of this study is to examine pitch-tracking patterns associated with capsulolabral injuries across 3 critical intervals: the season leading up to injury, the intervening offseason, and the period immediately preceding IL designation. Using a matched case-control design, key performance metrics such as velocity, release extension, release position, and arm angle were evaluated to determine whether short-term changes signal the onset or progression of capsulolabral pathology. Identifying these trends could provide the basis for early detection strategies and preventative measures aimed at reducing the risk or severity of capsulolabral injuries in professional pitchers.

METHODS

Study Design and Participant Selection

This study was deemed exempt as Not Human Subjects Research Under 45 CFR 46.102 by the institutional review board at Columbia University Medical Center, as no human participants were enrolled and all data were obtained from publicly available sources. A retrospective analysis was performed on MLB pitchers who underwent primary capsulolabral repair between March 1, 2017, and December 1, 2024.

Public injury databases used in previous reports, such as Fangraphs and Spotrac,^{11,28,29,33} were first utilized to identify potential cases. Public news reports and official team statements were then reviewed to confirm the injury type and surgical management. To ensure that included cases specifically involved capsulolabral pathology, we restricted inclusion to players for whom public reports explicitly described arthroscopic capsulolabral repair (eg, "labral repair," "capsular repair," or "capsule-labral repair"). Cases described only as nonspecific "shoulder

^{||}Address correspondence to Christopher S. Ahmad, MD, Department of Orthopedic Surgery, Columbia University Irving Medical Center–New York Presbyterian Hospital, 622 West 168th Street, PH-11, Room 1118, New York, NY 10032, USA (email: csa4@cumc.columbia.edu).

*University at Buffalo Department of Orthopaedics and Sports Medicine, Buffalo, New York, USA.

†Department of Orthopedic Surgery, Columbia University Irving Medical Center–New York Presbyterian Hospital, New York, New York, USA.

‡New York Institute of Technology College of Osteopathic Medicine, Old Westbury, New York, USA.

§Department of Orthopaedic Surgery, Wake Forest University School of Medicine, Winston-Salem, North Carolina, USA.

M. D. and M.A.M contributed equally to this paper.

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surgery,” “debridement,” or similar terms without clear reference to repair of the labrum or capsule were excluded. Three reviewers (M.D., N.F., and M.A.M.) independently verified classifications, with discrepancies resolved by consensus.

A total of 39 MLB players who underwent primary capsulolabral repair were identified during the study period. Pitchers were excluded if they (1) did not have 2 consecutive seasons of qualifying MLB data, defined as ≥ 100 tracked pitches in both the injury season (prior to IL placement) and the season immediately before, or (2) if surgery occurred during the offseason, between the end of the regular season and February 15 of the following year, which precludes reliable injury dating. This resulted in a cohort of 27 pitchers included in our final analysis.

Matched Control Selection

A 2:1 matched control cohort was assembled ($n = 54$) using methods consistent with prior case-control investigations of UCL injuries in professional pitchers.^{6,27,28,29,37} Demographic and player-context variables were obtained from Fangraphs.¹¹ Controls were first matched exactly to cases on season, playing role (starter vs reliever), and handedness. Within these groups, the 2 controls most similar to each case in age and preinjury workload (pitch count) were selected. Eligible controls were required to have no documented history of IL placement for elbow or shoulder reasons during the 2 years of data collection and no known history of shoulder or elbow surgery. The study also required controls to have ≥ 100 pitches thrown in both seasons analyzed.

Data Source, Variables, and Eligibility

Pitch-tracking data were obtained from Baseball Savant,³ which compiles tracking measurements from MLB’s Statcast. Our analysis considered 6 kinematic variables at the point of ball release: release speed (miles per hour [mph]), defined as the velocity of the baseball; spin rate (revolutions per minute), or the rotational velocity of the ball; release extension (feet), the distance from the pitching rubber to the point of release in the direction of home plate; arm angle (degrees), the horizontal angle between the pitcher’s throwing shoulder and the ball at release; and horizontal and vertical release position (feet), representing the ball’s spatial location at release, relative to the center of the pitching rubber (Figure 1). Statcast’s direct arm-angle metric was available beginning with the 2020 season; therefore, all arm-angle analyses were limited to the subset with available data (cases, $n = 18$; controls, $n = 36$), whereas models for other kinematic variables used the full analytic sample. Because Statcast reports horizontal release position with handedness-dependent sign conventions, absolute values were used to enable throwing-neutral comparisons.

Pitch types followed official Statcast labels, with analyses focusing on 7 specific categories: 4-seam fastball (FF), sinker (SI), slider (SL), curveball (CU), sweeper (ST),

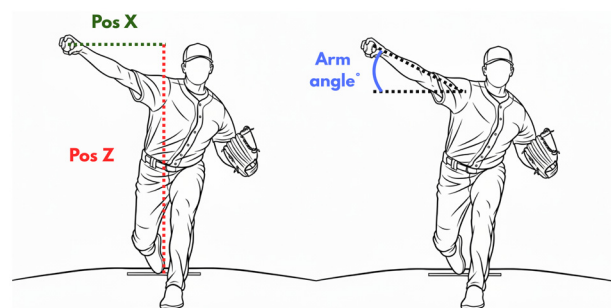


Figure 1. Illustration of pitch release kinematic measurements. Pos X represents the horizontal release position, a measure of the horizontal distance of the ball at release from the center of the rubber. Pos Z represents the vertical release position, which is the height of the ball compared with the rubber. Arm angle is the horizontal angle between the pitcher’s throwing shoulder and the ball at release. The authors used the generative AI tool Gemini (Google) for the purpose of creating a base illustrative silhouette of a pitcher. All scientifically relevant content, including schematic overlays and labels, were manually created and added by the authors.

changeup, and knuckle curve (KC). The initial data set comprised 161,979 pitches and 3347 pitcher-outings overall (season 1: cases 888, controls 1354; season 2: cases 423, controls 682) across all included pitchers. Within each outing, we computed the mean and pitch-to-pitch variability for each metric within a pitch type, with variability quantified as the within-outing standard deviation. The within-outing standard deviation was selected because it provides a direct measure of pitch-to-pitch consistency for a given pitcher during a single game, a metric used in prior biomechanical studies of pitching.^{15,26} A uniform eligibility rule was used across analyses: an outing contributed to a given pitch type only if it contained ≥ 4 pitches of that type; analysis-specific outing-count requirements (eg, ≥ 4 outings for baseline summaries and ≥ 6 outings for early versus late change analyses) are detailed below.

Study Timeline and Analysis Overview

For each case, the injury season (season 2) was defined as the MLB season in which IL placement occurred prior to capsulolabral repair, and the preinjury season (season 1) was the immediately preceding MLB season. The offseason interval (Δ Offseason) was defined as the period between the end of season 1 and the start of season 2. To identify potential risk factors and warning signs, 3 analyses were performed: (1) baseline case-control comparisons in season 1 using pitch type-specific mean and within-outing variability; (2) longitudinal change analyses using early versus late blocks across Δ S1 (within season 1), Δ Offseason (late season 1 to early season 2), and Δ S2 (within season 2); and (3) acute preinjury trend analyses modeling per-outing trajectories across the final 5 season 2 outings prior to IL

Study Timeline and Analysis Windows

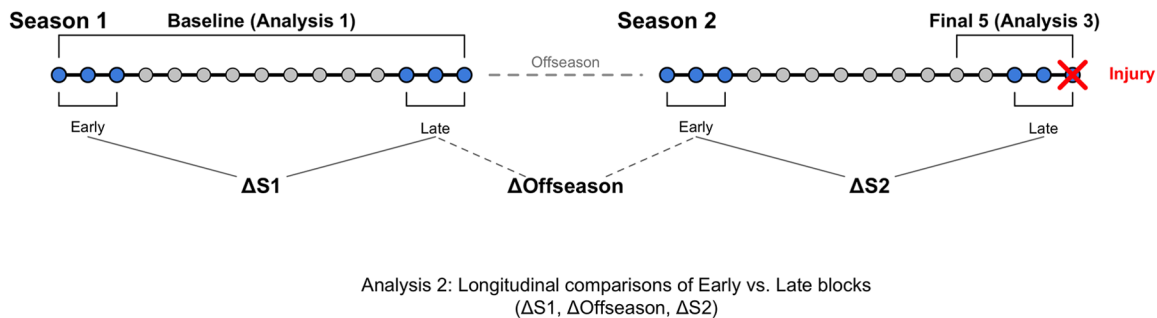


Figure 2. Study timeline and analytic framework. Schematic illustrating the definition of season 1 (preinjury season), season 2 (injury season), and the offseason interval ($\Delta\text{Offseason}$), along with the 3 primary analyses: (1) season 1 baseline comparisons; (2) longitudinal early versus late block comparisons across ΔS1 (within season 1), $\Delta\text{Offseason}$ (late season 1 to early season 2), and ΔS2 (within season 2); and (3) acute preinjury trajectories across the final 5 Season 2 outings prior to injured list placement.

placement. The study timeline and all 3 analyses are outlined in Figure 2.

Analysis 1—Season 1 Baseline by Individual Pitch Type

To assess baseline differences in season 1, 2 complementary analyses were conducted. First, to explore effects at a granular level, pitch type-specific differences were estimated separately for pitch types with >10 matched pairs contributing data (ie, ≥ 10 case-control pairs with ≥ 4 qualifying outings for that pitch type) (Appendix). This was accomplished by calculating the delta between a case's value and the mean value of its matched control(s) within each pair, followed by a 1-sample paired t test on these differences. Normality of the paired differences was assessed using the Shapiro-Wilk test prior to performing paired t tests. Second, a pooled linear model was fit to summarize the overall effect across all pitch types. For this model, each pitcher's metric values were first standardized (as z -scores) relative to the control group's mean and standard deviation for the corresponding pitch type. The model estimated the overall case-control difference with group (case vs control) as the primary predictor, while including fixed effects for both pitch type and matched-pair identification. To account for within-pitcher correlation from multiple pitch types, standard errors were clustered at the pitcher level.

Analysis 2—Longitudinal Change Using Primary FF and Primary Breaking Ball

To assess longitudinal changes, we selected each pitcher's primary FF (FF or SI) and primary breaking ball (SL, CU, ST, or KC), defined as the most-thrown pitch of its

type in season 1. To compare performance between the start and end of a given period, we created 2 discrete time points: an "early" block, representing the mean performance across the first 3 qualifying outings, and a "late" block, representing the mean across the last 3. We then fit linear mixed-effects models (LMMs) to these block-level data, with a random intercept for each pitcher. The model's time point $\times$ group interaction was the primary effect of interest, testing whether the change in performance from the early block to the late block was significantly different for cases compared with controls.

Analysis 3—Final 5 Outings (Season 2)

To assess trajectories immediately preceding injury, we isolated the final 5 outings for each pitcher in season 2. Using the same primary FF and breaking ball sets from analysis 2, we fit LMMs to the per-outing data. The time $\times$ group interaction tested for differential trends between cases and controls in the final outings leading up to the injury event, with a random intercept for each pitcher.

Analysis-Specific Sample Sizes and Eligibility

Because eligibility depended on within-season outing counts, sample sizes varied by analysis. For analysis 1 (season 1 baseline by individual pitch type), all 27 cases met eligibility for ≥ 1 FF pitch type and ≥ 1 breaking ball pitch type; controls followed the same eligibility rules. For analysis 2, ΔS1 required ≥ 6 season 1 outings contributing to the primary pitch; 27 cases and 54 controls qualified. The $\Delta\text{Offseason}$ window required ≥ 3 late S1 and ≥ 3 early S2 outings; 25 cases qualified. The ΔS2 window required ≥ 6 season 2 outings; 19 cases and 38 controls qualified. For analysis 3 (final 5 outings), eligibility required 5 season 2 outings; 22 cases qualified.

TABLE 1
Demographics and Baseline Characteristics of Labral Surgery Cases Versus Matched Controls^a

Variable	Cases (n = 27)	Controls (n = 54)	P
Age, y	28.6 ± 3.4	27.5 ± 2.9	.13
Height, cm	190.6 ± 7.2	188.0 ± 7.3	.13
Weight, kg	102.1 ± 9.8	96.1 ± 8.7	.01*
BMI	28.2 ± 3.5	27.2 ± 2.4	.20
Draft position	215.1 ± 291.9	201.4 ± 212.8	.85
Career MLB seasons	7.1 ± 3.5	6.2 ± 3.4	.29
Right-handed	77.8	77.8	≥.99
Starting pitcher	48.1	48.1	≥.99
International free agent	18.5	22.2	.78
Pitch count, season 1	1376 ± 903	1294 ± 942	.58

^aData are presented as mean ± SD or %. Season 1 represents the season prior to surgical intervention. *P* values are from paired comparisons. Asterisk shows statistical significance at **P* < .05. BMI, body mass index; MLB, Major League Baseball.

Because arm angle was only available since 2020, all arm-angle models used a reduced subset (cases, *n* = 18; controls, *n* = 36).

Statistical Analysis

All analyses were conducted in R (Version 4.3.1), with LMMs fit using the lmerTest package and clustered standard errors calculated with the clubSandwich package. A *P* value < .05 was considered statistically significant.

RESULTS

Demographics and Baseline Characteristics

Among the 27 pitchers who underwent primary capsulolabral repair and the 54 matched controls, matching achieved identical distributions of handedness (77.8% right-handed in both groups) and starting role (48.1% starters), with no between-group differences in age or workload at baseline (season 1 pitch count: 1376 ± 903 vs 1294 ± 942; *P* = .58).

Cases weighed more than controls (102.1 ± 9.8 kg vs 96.1 ± 8.7 kg; *P* = .01), whereas height, body mass index, draft position, MLB service time, and international free agent status were similar between groups (all *P* ≥ .13). Demographic data are summarized in Table 1.

Analysis 1—Baseline Risk Profile (Season 1)

By Individual Pitch Types. Looking within FFs and breaking balls, cases threw with higher velocity for both 4-seam FFs and SIs: +1.46 mph on FF (*P* = .009) and +1.57 mph on SI (*P* = .006). Variability signals were directionally consistent with tighter mechanics in cases: lower arm-angle variability on FF (*P* = .003), and lower release-speed variability on FF (*P* = .03) and SI (*P* = .05). Among breaking balls, SLs showed lower arm-angle variability

and a more medial release point on average (*P* = .001 and .02, respectively) (Appendix).

Pooled Across Pitch Types (Main Analysis). Standardizing pitcher-level values within pitch type and pooling with pitch-type and matched-pair fixed effects (standard errors clustered by pitcher), cases exhibited less within-outing variability in arm angle (β = −0.584 SD; 95% CI, −0.993 to −0.175; *P* = .005) and release speed (β = −0.370 SD; 95% CI, −0.637 to −0.102; *P* = .007), alongside higher means for release speed (β = +0.553 SD; 95% CI, 0.208 to 0.899; *P* = .002) and spin rate (β = +0.315 SD; 95% CI, 0.053 to 0.578; *P* = .02). Other pooled metrics were not significant. The complete data for the pooled results are presented in Table 2 and is visualized in Figure 3.

Analysis 2—Longitudinal Changes Preceding Surgery

For FFs, early- and offseason windows were largely null, with 2 exceptions: a small reduction in extension during $\Delta S1$ (*P* = .03) and a transient increase in release-speed variability during $\Delta \text{Offseason}$ (*P* = .04). The late-season window showed the clearest divergence: cases had a significant decline in mean release speed across $\Delta S2$ (*P* = .009) and slightly lower variability in horizontal release position (*P* = .03). Other FF metrics (arm angle, spin, release positions, extension variability) did not differ consistently between groups. See Table 3 for full windowed estimates.

For breaking balls, early- and offseason windows were mostly null. In $\Delta S1$, cases showed less lateral release variability (*P* = .04), while other within-season changes were not significant. Offseason differences across metrics were also not significant. By season 2, the clearest divergence emerged: cases demonstrated a larger decline in mean release speed across $\Delta S2$ (*P* = .003). No consistent group differences were observed for spin rate, arm angle, extension, or vertical release. See Table 4 for full windowed estimates on breaking balls.

TABLE 2
Season 1 Baseline Pooled Standardized Differences Across Pitch Types^a

Metric	Standard Difference (β)	95% CI	P
Arm angle, SD	-0.584	(-0.993 to -0.175)	.005**
Release speed, SD	-0.370	(-0.637 to -0.102)	.007**
Release spin rate, SD	0.363	(-0.017 to 0.744)	.06
Release pos X, SD	-0.065	(-0.519 to 0.389)	.78
Release extension, SD	-0.041	(-0.342 to 0.261)	.79
Release pos Z, SD	-0.027	(-0.300 to 0.246)	.85
Release speed, mean	0.553	(0.208 to 0.899)	.002**
Release spin rate, mean	0.315	(0.053 to 0.578)	.02*
Arm angle, mean	0.270	(-0.253 to 0.793)	.31
Release pos Z, mean	0.255	(-0.215 to 0.724)	.29
Release pos X, mean	-0.209	(-0.620 to 0.201)	.32
Release extension, mean	-0.137	(-0.455 to 0.181)	.40

^aStandardized case-control differences (β) in control-SD units for mean values and variability (SD) across 6 kinematic metrics. Results are from models with pitch-type and pair fixed effects and clustered SEs by pitcher. Positive values indicate higher means or variability in cases; negative values indicate lower values. Asterisks denote statistical significance at * $P < .05$ or ** $P < .01$.

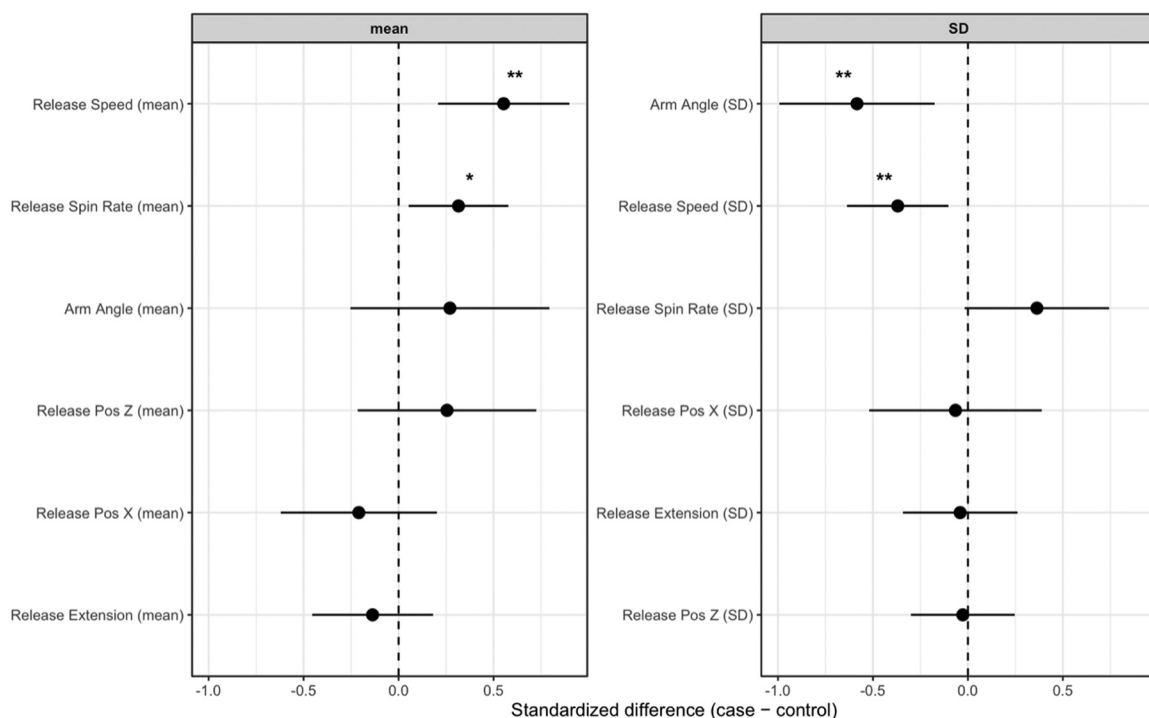


Figure 3. Season 1 baseline mechanics. Forest plot showing standardized case-control differences (β) for mean values and variability (SD) of 6 kinematic metrics. Values were standardized within pitch type relative to the control group and pooled using models with pitch-type and pair fixed effects; SEs were clustered by pitcher. Positive values indicate higher means or variability in cases, negative values indicate lower values. Error bars are 95% CIs. Asterisks denote significance at * $P < .05$ or ** $P < .01$. pos X, the horizontal release position, a measure of the horizontal distance of the ball at release from the center of the rubber; pos Z, the vertical release position, which is the height of the ball compared with the rubber.

Analysis 3—Final 5 Outings (Season 2)

Focusing on the last 5 appearances in season 2, cases showed a clear short-horizon velocity decline relative to

controls on both pitch FFs and breaking balls (Tables 5 and 6; Figure 4). On the primary FF, the per-outing slope difference was -0.195 mph/outing (95% CI, -0.301 to -0.090 ; $P < .001$), corresponding to roughly a 0.8-mph

TABLE 3
Windowed Deltas on Fastballs (mixed-effects)^a

Metric	$\Delta S1$ (n = 27)		$\Delta \text{Offseason}$ (n = 25)		$\Delta S2$ (n = 19)	
	Effect (95% CI)	P	Effect (95% CI)	P	Effect (95% CI)	P
Release speed, mean	-0.12 (-0.61 to 0.37)	.62	-0.39 (-0.98 to 0.19)	.18	-0.68 (-1.18 to -0.17)	.009**
Release pos X, mean	-0.05 (-0.21 to 0.10)	.48	0.10 (-0.07 to 0.26)	.25	-0.03 (-0.26 to 0.20)	.79
Release pos Z, mean	0.02 (-0.04 to 0.08)	.51	-0.04 (-0.11 to 0.04)	.30	0.03 (-0.05 to 0.11)	.51
Arm angle, mean	-1.65 (-3.42 to 0.12)	.07	-0.83 (-3.63 to 1.97)	.55	-0.38 (-2.79 to 2.02)	.75
Extension, mean	-0.08 (-0.15 to -0.01)	.03*	0.03 (-0.05 to 0.11)	.50	-0.01 (-0.08 to 0.06)	.72
Spin rate, mean	1.06 (-31.29 to 33.41)	.95	-24.13 (-58.83 to 10.57)	.17	-20.70 (-58.34 to 16.94)	.28
Release speed, SD	0.01 (-0.08 to 0.11)	.77	0.09 (0.01 to 0.18)	.04*	0.02 (-0.11 to 0.15)	.76
Release pos X, SD	-0.03 (-0.07 to 0.01)	.11	0.02 (-0.02 to 0.07)	.31	-0.02 (-0.03 to 0.00)	.03*
Release pos Z, SD	0.00 (-0.01 to 0.01)	.92	0.01 (-0.00 to 0.02)	.22	-0.00 (-0.01 to 0.01)	.50
Arm angle, SD	-0.16 (-0.48 to 0.16)	.32	0.15 (-0.18 to 0.47)	.37	-0.05 (-0.46 to 0.35)	.80
Extension, SD	-0.01 (-0.03 to 0.01)	.17	-0.00 (-0.02 to 0.02)	.91	0.01 (-0.03 to 0.04)	.73
Spin rate, SD	11.09 (-6.78 to 28.96)	.22	5.95 (-7.69 to 19.60)	.39	-9.45 (-25.05 to 6.14)	.23

^aCase-control differences in average pitch characteristics and within-outing variability for primary fastballs across 3 time frames: $\Delta S1$ (season 1 early vs late), $\Delta \text{Offseason}$ (season 1 late to season 2 early), and $\Delta S2$ (season 2 early vs late). Estimates are from linear mixed-effects models with a time point $\times$ group interaction term; effects are expressed in native units. Positive values indicate greater increases in cases relative to controls; negative values indicate greater decreases. Asterisks denote statistical significance at $*P < .05$ or $**P < .01$. pos X, the horizontal release position, a measure of the horizontal distance of the ball at release from the center of the rubber; pos Z, the vertical release position, which is the height of the ball compared with the rubber.

TABLE 4
Windowed Deltas on Breaking Balls (mixed effects)^a

Metric	$\Delta S1$ (n = 27)		$\Delta \text{Offseason}$ (n = 25)		$\Delta S2$ (n = 19)	
	Effect (95% CI)	P	Effect (95% CI)	P	Effect (95% CI)	P
Release speed, mean	-0.35 (-1.06 to 0.37)	.34	0.00 (-0.82 to 0.83)	>.99	-1.07 (-1.75 to -0.39)	.003**
Release pos X, mean	-0.06 (-0.22 to 0.10)	.45	0.07 (-0.10 to 0.23)	.43	-0.02 (-0.27 to 0.23)	.86
Release pos Z, mean	0.03 (-0.04 to 0.10)	.33	-0.03 (-0.10 to 0.05)	.51	0.02 (-0.04 to 0.08)	.53
Arm angle, mean	-1.44 (-3.95 to 1.07)	.25	-0.82 (-3.64 to 1.99)	.56	-0.15 (-1.85 to 1.56)	.86
Extension, mean	-0.07 (-0.15 to 0.01)	.10	0.07 (-0.02 to 0.16)	.11	-0.03 (-0.11 to 0.05)	.47
Spin rate, mean	-19.30 (-71.10 to 32.50)	.46	-17.18 (-76.25 to 41.89)	.56	31.55 (-18.15 to 81.24)	.21
Release speed, SD	0.03 (-0.16 to 0.21)	.78	0.03 (-0.16 to 0.21)	.76	0.05 (-0.13 to 0.23)	.61
Release pos X, SD	-0.05 (-0.10 to -0.00)	.04*	0.01 (-0.03 to 0.06)	.56	0.01 (-0.01 to 0.03)	.22
Release pos Z, SD	0.01 (-0.00 to 0.02)	.09	-0.01 (-0.02 to 0.01)	.32	0.00 (-0.01 to 0.01)	.64
Arm angle, SD	-0.29 (-0.71 to 0.13)	.17	0.08 (-0.36 to 0.52)	.72	0.07 (-0.37 to 0.51)	.74
Extension, SD	-0.02 (-0.06 to 0.01)	.18	0.00 (-0.03 to 0.03)	.94	0.03 (-0.01 to 0.07)	.11
Spin rate, SD	-25.23 (-57.36 to 6.89)	.12	-23.01 (-71.48 to 25.46)	.35	-9.89 (-27.78 to 8.00)	.27

^aCase-control differences in average pitch characteristics and within-outing variability for primary breaking balls across 3 time frames: $\Delta S1$ (season 1 early vs late), $\Delta \text{Offseason}$ (season 1 late to season 2 early), and $\Delta S2$ (season 2 early vs late). Estimates are from linear mixed-effects models with a time point $\times$ group interaction term; effects are expressed in native units. Positive values indicate greater increases in cases relative to controls; negative values indicate greater decreases. Asterisks denote statistical significance at $*P < .05$ or $**P < .01$. pos X, the horizontal release position, a measure of the horizontal distance of the ball at release from the center of the rubber; pos Z, the vertical release position, which is the height of the ball compared with the rubber.

drop from the first to the fifth outing. On the primary breaking ball, the decline was steeper: -0.298 mph/outing (95% CI, -0.445 to -0.150; $P < .001$), or about 1.2 mph over the same span. No other mean kinematic measures (arm angle, extension, spin rate, or lateral/vertical release position) showed significant between-group trends during this window (all $P \geq .17$). Analyses included 22 eligible cases with matched controls.

DISCUSSION

The principal findings of this matched case-control study of MLB pitchers undergoing arthroscopic capsulolabral repair reveal a distinct pitch profile associated with these severe shoulder injuries. First, at baseline in the season prior to injury, pitchers who would eventually require surgery exhibited a unique kinematic profile characterized by

TABLE 5
Fastballs—Linear Mixed-Effects Model Over Final 5 Outings^a

Metric	Effect (95% CI)	P
Release speed, mean	−0.195 (−0.301 to −0.090)	<.001**
Release pos X, mean	−0.008 (−0.033 to 0.016)	.49
Release pos Z, mean	0.002 (−0.011 to 0.015)	.77
Arm angle, mean	0.157 (−0.117 to 0.430)	.26
Extension, mean	−0.001 (−0.019 to 0.016)	.89
Spin rate, mean	−2.470 (−12.520 to 7.580)	.63

^aThe reported effect is the time × group interaction (cases–controls) in native units for each measure. Negative values indicate steeper declines in cases. Entries are estimates with 95% CIs and *P* values; eligibility required 5 outings (cases, *n* = 22). Asterisks denote statistical significance at *P* < .01. pos X, the horizontal release position, a measure of the horizontal distance of the ball at release from the center of the rubber; pos Z, the vertical release position, which is the height of the ball compared with the rubber.

TABLE 6
Breaking Balls—Linear Mixed-Effects Model Over Final 5 Outings^a

Metric	Effect (95% CI)	P
Release speed, mean	−0.298 (−0.445 to −0.150)	<.001**
Release pos X, mean	−0.007 (−0.032 to 0.018)	.59
Release pos Z, mean	0.009 (−0.005 to 0.024)	.20
Arm angle, mean	0.176 (−0.135 to 0.488)	.27
Extension, mean	−0.002 (−0.025 to 0.021)	.88
Spin rate, mean	−9.341 (−22.690 to 4.008)	.17

^aThe reported effect is the time × group interaction (cases–controls) in native units for each measure. Negative values indicate steeper declines in cases. Entries are estimates with 95% CIs and *P* values; eligibility required 5 outings (cases, *n* = 22). Asterisks denote statistical significance at *P* < .01. pos X, the horizontal release position, a measure of the horizontal distance of the ball at release from the center of the rubber; pos Z, the vertical release position, which is the height of the ball compared with the rubber.

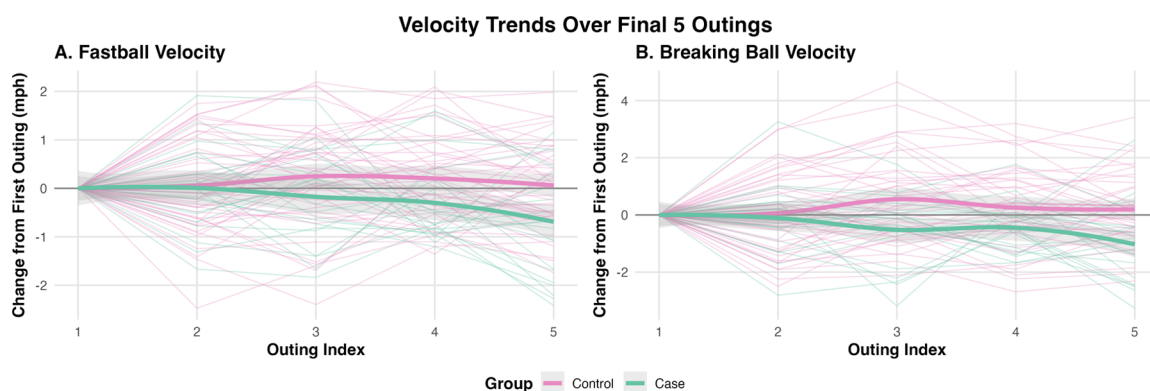


Figure 4. Mean release speed for cases and matched controls across each pitcher's last 5 appearances in Season 2 for (A) the primary fastball and (B) the primary breaking ball. Values are normalized to the first outing of the sequence. Bold lines indicate Locally Estimated Scatterplot Smoothing group trends, while thin lines represent individual pitcher trajectories. Cases are shown in teal and matched controls in pink.

significantly higher mean release speed and spin rate, and notably, significantly lower within-outing variability in arm angle and release speed. Second, longitudinal windows were mostly quiet until late season 2, when cases

demonstrated a significant decline in velocity on both FFs and breaking balls. Third, this velocity degradation was found to accelerate sharply in the final 5 outings immediately preceding their placement on the IL. A

secondary demographic finding was that cases had a significantly higher body weight than their matched controls.

The baseline findings suggest a novel biomechanical risk profile for capsulolabral injury rooted in a combination of high intensity and high consistency throwing. The observation that cases threw harder and with more spin is consistent with foundational biomechanical principles: generating elite velocity requires and imparts greater kinetic forces upon the shoulder's static stabilizers, including the labrum and capsule.^{1,23,35} This is consistent with prior large-scale investigations identifying pitch velocity and spin rate as key predictors of elbow injury.^{17,29,30,36} To our knowledge, this study is the first to demonstrate that this specific combination of high-intensity metrics is associated with a surgically confirmed cohort of pitchers with capsulolabral pathology.

The more novel finding from our baseline analysis was the significantly lower kinematic variability among cases. Pitchers who ultimately required surgery demonstrated a more repeatable delivery, with less deviation in arm angle and release speed on a pitch-to-pitch basis. At first glance, this appears to contradict the findings of Manzi et al,²⁶ whose motion-capture analysis found that professional pitchers with higher consistency actually experienced lower peak shoulder distraction forces. However, we propose that this finding highlights a crucial distinction between acute peak forces and chronic cumulative stress. This suggests that while mechanical consistency may be protective against peak joint distraction loads, the trade-off for the shoulder may be the insidious, cyclic strain on the static stabilizers, ultimately predisposing to fatigue failure of the labrum and capsule.

The finding of decreased variability as a potential risk factor for capsulolabral surgery stands in direct contrast to studies of UCL injury, which have suggested that increased variability in release location may be a risk factor for reconstruction.^{10,44,45} This apparent contradiction suggests a fundamental divergence in the injury mechanisms of the shoulder and elbow. The UCL is a ligamentous structure that primarily resists valgus torque in a single plane; excessive kinematic variability may represent a breakdown in the kinetic chain, leading to unpredictable force spikes that acutely overload the ligament.^{24,44} This concept ties into the adaptive change theory,^{8,20,21} which proposes that repeated mechanical loading can remodel the olecranon. Over time, this may lead to features such as osteophyte formation, which in turn can provide a stabilizing effect on the elbow joint. Variability in pitching mechanics may prevent these adaptive changes from occurring in the elbow. To date, this adaptive change theory has not been described as a mechanism influencing shoulder stability. In a hypermobile and multiaxial ball-and-socket joint, whose stability is heavily dependent on the dynamic coordination of the rotator cuff and the static restraint of the capsulolabral complex, formation of adaptive protective changes may not play a significant role influencing stability.^{47,48} It is possible that for the shoulder, a hyperconsistent delivery, while performatively advantageous, concentrates repetitive microtrauma onto focal areas of the labrum and capsule, leading to attritional

wear and faster eventual failure. It is also possible that the larger variability in the control group is a result of not throwing 100% for every pitch, and thus in turn these pitchers are "pacing" themselves while experiencing potentially lower joint loads and less fatigue.

The windowed analysis was an attempt to determine if specific kinematic changes preceded injury by isolating distinct time periods: within the baseline season ($\Delta S1$), the offseason ($\Delta \text{Offseason}$), and the injury season ($\Delta S2$). The findings from the first 2 windows were largely null, with only a few isolated significant results that did not show a consistent pattern and may represent statistical noise from multiple comparisons. However, the analysis of $\Delta S2$ revealed a clear and significant trend: cases experienced a greater decline in mean release speed on both FFs and breaking balls compared with controls. This suggests that while a pitcher's baseline mechanical signature represents a chronic, underlying risk profile, the detectable degradation in performance is a more acute phenomenon that manifests closer to the time of clinical injury.

Focusing on the critical window immediately before IL placement, our analysis of the final 5 outings reinforces the velocity decline as the most robust signal of injury. Although further prospective studies are needed to clarify the clinical significance of these findings and determine whether specific preventive actions are warranted, a sustained decline in velocity over even a small sequence of pitches may reasonably prompt consideration of pitcher removal. Sustained velocity loss has been associated with late fatigue and increased injury risk.²⁷ Such declines are often accompanied by mechanical inconsistency and late compensatory biomechanical patterns, reflecting kinetic chain disruption that can increase valgus loading at the elbow and deceleration stresses on the posterior capsulolabral complex.^{9,13} This finding contrasts with prior work on shoulder injuries that did not detect significant preinjury velocity changes, a difference that may be attributable to our specific focus on capsulolabral pathology and our rigorous matched-control design.⁴² We posit that velocity is the most sensitive and objective indicator of declining physical capacity. A pitcher, through proprioception and skill, may be able to maintain the gross kinematic markers of his delivery—such as arm angle and release position—but cannot overcome the pain, instability, or neuromuscular inhibition from a developing structural injury to generate maximal force and thus velocity. Overall, this is among the first studies to apply a windowed analysis to preinjury shoulder kinematics and by isolating a preinjury mechanical signature that precedes eventual velocity decline. Our work offers a potential pathway to move beyond reactive surgical intervention and toward proactive monitoring, addressing a critical gap in the literature that has historically focused more on postsurgical outcomes in shoulder injuries,⁴¹ than on primary prevention. Based on these data, prevention efforts may be better oriented toward identifying baseline risk profiles rather than relying solely on the acute preinjury window. Pitchers who pair higher output (velocity and spin) with highly repeatable release mechanics may represent a subgroup in whom repetitive, focal loading of the capsulolabral complex

warrants closer longitudinal surveillance. Although mechanical consistency is typically advantageous for performance, our findings raise the possibility that an extreme “hyperconsistent” profile, particularly in arm angle and release speed, may reflect concentrated repetitive stress. Given the clear performance value of velocity, spin, and repeatable mechanics, these observational findings should not be interpreted as advising pitchers to reduce output or intentionally increase mechanical variability; rather, future work should determine how pitch-to-pitch variability relates to both performance and injury risk and whether any actionable monitoring or modulating strategies are feasible without sacrificing competitive outcomes.

The finding that cases had significantly greater body mass than matched controls also warrants consideration. A 2024 machine learning analysis by Oeding et al³⁶ studied all MLB players with shoulder or elbow injuries and found increased body weight to be the strongest player demographic variable in their risk models. Other UCL-related risk factor studies have made similar findings.^{6,16} Our result provides further evidence for this association, specifically within the context of labral injuries. This may simply be another characteristic of the high-output pitcher phenotype, as greater body mass can contribute to generation of higher pitch velocities.⁴⁹ However, further research is needed to determine if body mass is an independent causal factor, perhaps by altering the kinetic chain or increasing inertial loads on the joint, or merely a covariate of the ability to throw with high velocity.

Limitations

This study is not without limitations. First, because it is retrospective and dependent on publicly available information, it cannot establish causality. Although these public injury databases have been established previously, they are subject to incomplete or inaccurate characterization of operative diagnoses and procedures.¹⁹ We were also unable to access or cross-reference the MLB Health and Injury Tracking System (HITS), which contains detailed surgical documentation; however, because HITS is deidentified, it cannot be linked to publicly identifiable players or integrated with pitch-level Statcast data, limiting its utility for studies of this design.³² As a result, important surgical details, such as tear location, degree of bone loss, and number or type of anchors used, were not available and may have influenced our findings. Despite these limitations, MLB injury reporting has expanded substantially in recent years, and teams increasingly disclose injuries requiring surgery and prolonged absence, making this a reasonably sized case-control cohort that compares favorably with prior studies in this area. In addition, the case count was modest, and several analyses were further constrained by pitch type–specific eligibility requirements and incomplete availability of certain Statcast variables (eg, arm angle since 2020). Accordingly, absence of statistically significant differences for certain metrics should be viewed as inconclusive rather than evidence of no relationship.

The timing of injuries was approximated using IL designations and surgical reports, which may not perfectly align

with the actual onset of symptoms. Although offseason surgeries were excluded to reduce this uncertainty, some relevant biomechanical alterations may still have occurred outside of the analyzed intervals. In addition, because the longitudinal analyses used discrete early versus late blocks and a short preinjury window, we did not model continuous kinematic trajectories across the entirety of season 2. Earlier warning signs—potentially more amenable to rest or nonoperative management—may therefore have been missed. Furthermore, Statcast provides only external performance and kinematic measures; it does not capture internal joint forces, muscular activity, or ligamentous loading. Therefore, the observed mechanical patterns should be interpreted as associations with injury risk rather than direct measures of tissue stress. Finally, it is possible that some pitchers in the control group had undiagnosed, asymptomatic labral or other pathologies, as imaging studies have identified a high prevalence of such findings in the shoulders of asymptomatic elite throwers.²⁵ The presence of these subclinical conditions in the control group would likely bias our results toward the null, meaning the true risk factors we identified may be stronger than what we reported. Despite these limitations, this study contributes substantially to our understanding of risk factors for capsulolabral injuries in MLB pitchers.

CONCLUSION

MLB pitchers who required arthroscopic capsulolabral repair demonstrated a pitch profile characterized by higher velocity and spin coupled with reduced pitch-to-pitch variability. Progressive FF and breaking ball velocity decline was the clearest marker of impending injury, highlighting a measurable preinjury signature that may inform early detection and intervention.

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ORCID iDs

Morgan R. Dillon  <https://orcid.org/0009-0009-5032-316X>
 Michael A. Mastroianni  <https://orcid.org/0000-0001-8332-5131>
 Nicholas Frappa  <https://orcid.org/0009-0007-0623-2864>
 Danil Chernov  <https://orcid.org/0009-0001-1310-623X>
 Andrew J. Luzzi  <https://orcid.org/0000-0001-9714-3873>
 Matthew G. Alben  <https://orcid.org/0000-0001-7398-4088>
 James Muscat  <https://orcid.org/0009-0003-2133-9249>
 Kristen F. Nicholson  <https://orcid.org/0000-0002-7144-5715>

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APPENDIX: SEASON 1—PAIRED CASE-CONTROL DIFFERENCES BY PITCH TYPE

Season 1—Paired case-control differences by pitch type

 Δ = case – mean(controls) within matched pair; pitch types sorted by contributing pairs (descending)

	Paired Δ (case – ctrl)	95% CI	P-value
Four-seam Fastball (FF)—Matched pairs contributing=22; Contributing pitchers: cases=22, ctrls=38; Contributing outings: cases=543, ctrls=844			
Release Speed (mean)	1.460	[0.397, 2.523]	.009**
Release Pos X (mean)	-0.146	[-0.520, 0.227]	.424
Release Pos Z (mean)	0.217	[-0.019, 0.452]	.069
Release Extension (mean)	-0.037	[-0.234, 0.160]	.702
Arm Angle (mean)	3.508	[-3.778, 10.794]	.317
Release Spin Rate (mean)	49.610	[-28.360, 127.580]	.200
Release Speed (SD)	-0.094	[-0.175, -0.013]	.025*
Release Pos X (SD)	-0.009	[-0.062, 0.045]	.742
Release Pos Z (SD)	-0.005	[-0.012, 0.002]	.122
Release Extension (SD)	0.003	[-0.013, 0.018]	.713
Arm Angle (SD)	-0.411	[-0.659, -0.163]	.003**
Release Spin Rate (SD)	-2.177	[-14.355, 10.001]	.714
Slider (SL)—Matched pairs contributing=17; Contributing pitchers: cases=17, ctrls=26; Contributing outings: cases=426, ctrls=471			
Release Speed (mean)	1.011	[-1.237, 3.259]	.354
Release Pos X (mean)	-0.474	[-0.864, -0.084]	.020*
Release Pos Z (mean)	0.254	[-0.144, 0.652]	.194
Release Extension (mean)	-0.102	[-0.325, 0.121]	.345
Arm Angle (mean)	9.554	[-2.926, 22.034]	.119
Release Spin Rate (mean)	152.705	[-13.720, 319.129]	.070
Release Speed (SD)	-0.041	[-0.206, 0.125]	.611
Release Pos X (SD)	0.001	[-0.042, 0.043]	.976
Release Pos Z (SD)	-0.006	[-0.018, 0.007]	.339
Release Extension (SD)	-0.001	[-0.023, 0.020]	.919
Arm Angle (SD)	-0.413	[-0.620, -0.206]	.001**
Release Spin Rate (SD)	33.064	[-6.991, 73.119]	.099
Sinker/Two-seam (SI)—Matched pairs contributing=14; Contributing pitchers: cases=14, ctrls=21; Contributing outings: cases=347, ctrls=398			
Release Speed (mean)	1.570	[0.526, 2.613]	.006**
Release Pos X (mean)	-0.263	[-0.668, 0.142]	.184
Release Pos Z (mean)	0.138	[-0.238, 0.515]	.442
Release Extension (mean)	-0.242	[-0.531, 0.048]	.094
Arm Angle (mean)	3.922	[-11.070, 18.914]	.556
Release Spin Rate (mean)	22.520	[-78.227, 123.267]	.637
Release Speed (SD)	-0.119	[-0.238, -0.001]	.049*
Release Pos X (SD)	-0.066	[-0.150, 0.017]	.111
Release Pos Z (SD)	0.001	[-0.020, 0.021]	.950
Release Extension (SD)	0.015	[-0.017, 0.046]	.333
Arm Angle (SD)	0.038	[-0.574, 0.651]	.886
Release Spin Rate (SD)	-7.607	[-22.121, 6.907]	.278
Changeup (CH)—Matched pairs contributing=13; Contributing pitchers: cases=13, ctrls=21; Contributing outings: cases=274, ctrls=365			
Release Speed (mean)	0.292	[-1.570, 2.153]	.739
Release Pos X (mean)	-0.064	[-0.695, 0.566]	.828
Release Pos Z (mean)	-0.068	[-0.370, 0.234]	.634
Release Extension (mean)	-0.088	[-0.386, 0.210]	.530
Arm Angle (mean)	-0.870	[-11.536, 9.796]	.848
Release Spin Rate (mean)	-51.436	[-182.993, 80.122]	.411
Release Speed (SD)	0.057	[-0.148, 0.263]	.556
Release Pos X (SD)	0.021	[-0.044, 0.086]	.496
Release Pos Z (SD)	-0.005	[-0.021, 0.011]	.473
Release Extension (SD)	-0.009	[-0.040, 0.023]	.553
Arm Angle (SD)	-0.155	[-0.587, 0.277]	.414
Release Spin Rate (SD)	2.853	[-37.276, 42.982]	.879